

Geometric mass balance on the overlap graph and a two-anchor reduction of the coincidence conjecture*

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1 Setup

Let σ be a primitive irreducible Pisot substitution on $\mathcal{A} = \{1, \dots, k\}$ with incidence data $A_{ab} = |\sigma(a)|_b$, Perron value $\beta > 1$ of degree k , and tile lengths $L = (L_1, \dots, L_k) > 0$ with $AL = \beta L$. Irreducibility of the characteristic polynomial makes $L_1, \dots, L_k$ linearly independent over $\mathbb{Q}$. A *genuine overlap* is a triple $s = (i, \delta, j)$ with $\delta \in \mathbb{Z}^k$, $t(s) = \langle \delta, L \rangle$, such that the tiles $[0, L_i)$ and $[t, t + L_j)$ intersect with nonempty interior; write $o(s)$ for the intersection and $w(s) = |o(s)| > 0$. Inflation sends s to the cells of the common refinement of the $\sigma(i)$ - and $\sigma(j)$ -subdivisions restricted to $\beta \cdot o(s)$; each cell is a genuine overlap (i', δ', j') with $\delta' = A^\top \delta + \text{pp}_i(p) - \text{pp}_j(q)$ for prefix-Parikh vectors pp . A *coincidence* is a state $(i, 0, i)$. Overlap coincidence is equivalent to pure discreteness of the suspension tiling dynamics [10, 3]. The state space reachable from return-vector seeds is finite (Meyer-set arguments; cf. [3]), so no balanced-pair termination input is used anywhere below.

2 Mass balance

Theorem 2.1 (Partition identity). *For every genuine overlap s , counting children with multiplicity, $\beta w(s) = \sum_{c \text{ child of } s} w(c)$.*

Proof. $\beta \cdot [0, L_i)$ is partitioned (up to endpoints) by the sub-tile intervals of $\sigma(i)$, and $\beta \cdot [t, t + L_j)$ by those of $\sigma(j)$. The common refinement of the two partitions, restricted to $\beta \cdot o(s)$, is an interval partition of $\beta \cdot o(s)$ whose cells of nonempty interior are exactly the child overlaps. Sum lengths. $\square$

Let C be a recurrent noncoincident strongly connected component (SCC) of the genuine overlap graph, N_C its internal child-count matrix (nonnegative, integer, irreducible), $w|_C > 0$, and $\text{leak}(s) \geq 0$ the total measure of children of s outside C . Theorem 2.1 restricts to

$$\beta w = N_C w + \text{leak} \quad \text{on } C. \quad (1)$$

Theorem 2.2 (Ceiling and leak dichotomy). *$\rho(N_C) \leq \beta$, with equality iff $\text{leak} \equiv 0$ on C .*

*Companion section to the consolidated PSC manuscript (v13 lineage). All spectral claims herein are conditional content in the sense of v13: nothing in this note proves the Pisot substitution conjecture; it reorganizes the open core.

Proof. $N_C w \leq \beta w$ entrywise with $w > 0$ gives $\rho(N_C) \leq \beta$ (Collatz–Wielandt). If $\text{leak} \equiv 0$ then w is a positive eigenvector, hence the Perron eigenvector, and $\rho(N_C) = \beta$. If $\text{leak} \not\equiv 0$, pair (1) with the left Perron eigenvector $u > 0$ of N_C : $(\beta - \rho(N_C))\langle u, w \rangle = \langle u, \text{leak} \rangle > 0$. $\square$

Theorem 2.3 (Exact gap formula). $\beta - \rho(N_C) = \frac{\langle u, \text{leak} \rangle}{\langle u, w \rangle}$.

Remark 2.4. *Theorem 2.1 is elementary and implicit in the framework of [3]; we record it because (1) and Theorems 2.2–2.3 follow from it in two lines each, and because the identical pairing applied to the balanced-pair mass-balance identity of the consolidated manuscript yields the same gap formula on the symbolic side. The spectral gap of a recurrent overlap SCC is its eigenvector-weighted coincidence-production rate. For a closed C (no noncoincident child outside C), leak is coincidence mass only, so $\rho(N_C) < \beta$ iff some state of C has a coincidence child. The open core of the conjecture is therefore the same statement on both the symbolic and geometric sides: a closed recurrent noncoincident component must leak.*

Theorem 2.5 (Size floor for the counterexample shape). *If C is closed, recurrent, noncoincident with $\text{leak} \equiv 0$, then $\beta \in \text{spec}(N_C)$, hence the characteristic polynomial of A divides that of N_C and $|C| \geq k$. Moreover, writing $w = VL$ with the integer matrix V determined by the four overlap cases $v_s \in \{e_j, e_i - \delta_s, \delta_s + e_j, e_i\}$, linear independence of the L_a upgrades (1) to the integer intertwining $N_C V = V A$ with V injective.*

Proof. $N_C w = \beta w$ with $w \neq 0$, $w \in \mathbb{Q}(\beta)^{|C|}$, exhibits β as an eigenvalue of an integer matrix; the minimal polynomial of β is the (irreducible) characteristic polynomial of A , of degree k . Independence of the L_a converts $N_C V L = V A L$ into $N_C V = V A$; since $\mathbb{Q}^k$ is a simple $\mathbb{Q}[A]$ -module, $V \neq 0$ is injective. $\square$

3 The two anchors

Call $s = (i, \delta, j)$ *prefix-aligned* if $\delta = 0$ (the tiles share their left endpoint) and *suffix-aligned* if $\delta = e_i - e_j$ (then $t = L_i - L_j$: the tiles share their right endpoint). The prefix-aligned children of a prefix-aligned state are the equal-prefix-Parikh cells (the Barge–Diamond graph [7, 1]); the suffix-aligned children of a suffix-aligned state are the equal-suffix-Parikh cells, i.e. the Barge–Diamond graph of the reversed substitution $\tilde{\sigma}$ (same incidence matrix).

Theorem 3.1 (Prefix channel). *Let C be closed, recurrent, noncoincident, containing a prefix-aligned state $(i, 0, j)$. If (i, j) satisfies strong coincidence (some $\sigma^n(i)$, $\sigma^n(j)$ share a letter at equal prefix-Parikh), then C produces a coincidence child.*

Theorem 3.2 (Suffix channel). *Same statement with a suffix-aligned anchor $(i, e_i - e_j, j)$ and strong coincidence at equal suffix-Parikh (equivalently, strong coincidence of $\tilde{\sigma}$ for (i, j)).*

Proof of both. Anchor the two inflation towers at the common endpoint. For each level $m \leq n$, the sub-tile pairs containing the depth- n coincidence tile both contain its interior, hence form genuine overlaps; by closedness this nested chain lies in C until its final step exhibits a coincidence child of a state of C . $\square$

Remark 3.3 (γ -channel). *The proofs of Theorems 3.1–3.2 use only that the two towers are anchored at a common point realized by matching control points. The same nested-chain argument*

proves: if a closed recurrent noncoincident SCC contains a state (i, δ, j) with $\langle \delta, L \rangle = c_j - c_i$ for an admissible control-point system (c_a) arising from a tile map of some power of the substitution, and (i, j) satisfies strong coincidence with respect to that system, then the SCC produces. Theorems 3.1 and 3.2 are the instances given by the first-letter and last-letter tile maps; we single them out because Conjecture 3.7 asserts that these two anchors always occur, whereas [2] manufactures the anchor per trapped component (Remark 3.9).

Lemma 3.4 (Cycle pinning). *A directed cycle of the overlap graph with edge digit word $(d_1, \dots, d_n) \subset \mathbb{Z}^k$ carries the uniquely determined integer offset orbit $\delta_1 = (I - (A^\top)^n)^{-1} \sum_{i=1}^n (A^\top)^{n-i} d_i$, since $\det(I - (A^\top)^n) = \prod_m (1 - \beta_m^n) \neq 0$ for Pisot β .*

Lemma 3.5 (Shared cut). *Let s belong to a closed recurrent noncoincident SCC C and suppose a refinement window of s contains an interior cut point common to both grids. Then C contains a prefix- or suffix-aligned state.*

Proof. At a common cut both grids end one sub-tile and begin another; the cell ending there is right-aligned, the cell beginning there is left-aligned. A coincident cell is both, and its endpoints are again common cuts, so the maximal coincidence run through the cut is bounded by common cuts whose outside-adjacent cells are aligned and noncoincident, hence in C by closedness. If the run is flush against one window edge, the aligned noncoincident cell at its boundary on the *other* side serves; a run flush against both edges fills the window and leaves s without noncoincident children, contradicting recurrence. $\square$

Theorem 3.6 (Anchors from production). *Every producing closed recurrent noncoincident SCC contains a prefix- or suffix-aligned state.*

Proof. A coincidence cell is aligned on both sides, so its endpoints are common cuts; apply Lemma 3.5. $\square$

By Theorem 3.6, anchoring is automatic for producing components; the conjectural content concerns only the trapped shape of Theorems 2.2 and 2.5:

Conjecture 3.7 (Trapped anchoring, H3'). *Every closed recurrent noncoincident SCC with $\text{leak} \equiv 0$ contains a prefix-aligned or a suffix-aligned state.*

In a $\text{leak} \equiv 0$ component no coincidence cells exist, so by Lemma 3.5 *any* common interior cut anchors it immediately. Under the hypotheses of Theorem 3.8 a surviving counterexample must therefore keep its two cut grids totally disjoint at every refinement level, in windows growing like β^n — an alignment-avoidance condition at all scales, layered on the size floor and intertwining of Theorem 2.5. The Pisot property places $\beta^n t$ at bounded distance from the cut module; “bounded” versus “zero” is the residue, in contact with Property (F)-type finiteness statements of β -numeration.

Theorem 3.8 (Two-anchor reduction). *If σ and its reversal $\tilde{\sigma}$ satisfy the strong coincidence condition and Conjecture 3.7 (trapped anchoring) holds for σ , then every closed recurrent noncoincident SCC produces; consequently overlap coincidence holds and the tiling dynamical system of σ has pure discrete spectrum.*

Proof. Theorems 3.1–3.2 on the anchor supplied by Conjecture 3.7; every closed recurrent SCC leaks, and nonclosed recurrent SCCs leak by definition through their escapes together with Theorem 2.2; conclude by [10, 3]. $\square$

Remark 3.9 (Calibration against the literature). *Akiyama–Lee [4] prove overlap coincidence $\Rightarrow$ (generalized) strong coincidence when the height group is trivial, which holds for one-dimensional irreducible Pisot substitutions [8, 9]. Akiyama [2, Thm. 3.2, Cor. 3.3] proves: multiple strong coincidence of level n (strong coincidence for all admissible control-point systems of Ω^n with $\Lambda - \Lambda \subset \langle G \rangle$) is equivalent to overlap coincidence for the suspension tiling, with n bounded by the number of overlap classes. His proof shows the hypothetical trapped component has spectral radius $|\det Q| = \beta$ — precisely the leak $\equiv 0$ shape of Theorems 2.2 and 2.5 — and then constructs a tile map ([2, Lem. 3.5]) whose control points match the offset of a mixed-letter state of the trapped component: the anchor is moved to the offset. Thus γ -generalized anchoring of the trapped component is unconditional in [2], and the residue of his reduction is strong coincidence for the manufactured systems. Prefix- and suffix-strong coincidence are the control-point systems of the first-letter and last-letter tile maps (by Lemma 3.4, these are exactly the tile maps whose cycles carry zero digit corrections, pinning δ to the two endpoint anchors). Theorem 3.8 is therefore a sharpening proposal for the minimization question of [2, Rem. 3.4] (“can we take $n = 1$?”): conditional on Conjecture 3.7, the two endpoint systems suffice. Conjecture 3.7 is strictly stronger than what [2] needs — its failure shape, a closed component whose pinned offsets avoid $\{0\} \cup \{L_i - L_j\}$, is exactly what his construction routes around, and exactly what the censuses below never exhibit.*

4 The ℓ -fixed reduction of Conjecture 3.7

For a state of a leak $\equiv 0$ component every refinement cell is noncoincident and in the component, so the leftmost- and rightmost-cell maps are self-maps of the component; being eventually periodic on a finite set, *every trapped component contains an ℓ -cycle and an r -cycle*, and a period- p ℓ -cycle state is ℓ -fixed for σ^p (the leftmost cell of the level- p refinement is the p -fold leftmost cell). Trappedness persists under powers and descendant cones are power-independent.

Lemma 4.1 (Telescoping). *Let $s^* = (i^*, \delta^*, j^*)$ be ℓ -fixed with $t^* > 0$, so that $\sigma(j^*)$ begins with j^* , $\sigma(i^*) = u^* i^* v^*$, and $(A^\top - I)\delta^* = \pi(u^*)$ (the matrix is invertible since 1 is not an eigenvalue of A ; $v^* \neq \varepsilon$ is forced by $t^* < L_{i^*}$). Then, exactly, $(A^\top)^n \delta^* = \delta^* + \pi(Q_n)$ where $Q_n = \sigma^{n-1}(u^*) \cdots \sigma(u^*) u^*$ is the prefix of $\sigma^n(i^*)$ preceding the tracked central occurrence of i^* .*

Consequently a level- n shared cut for s^* is the *static* membership $\delta^* \in \{\pi(a) - \pi(v)\}$ over factors a anchored at the central occurrence and prefixes v of $\sigma^n(j^*)$; in the limit, over prefixes of the two σ -fixed half-tilings $\omega_I = i^* v^* \sigma(v^*) \cdots$ and $\omega_J = \lim \sigma^n(j^*)$ at offset t^* . This is the algebraic coincidence of Lee [6] localized at the single vector δ^* ; an unconditional proof would prove the Pisot conjecture.

Theorem 4.2 (ℓ -fixed reduction). *Conjecture 3.7 holds if for every $p \geq 1$ the finite statement $Q(p)$ holds: every in-window, integral, ℓ - or r -fixed state of σ^p contains an anchor or a coincidence in its descendant cone.*

Two unconditional fences accompany this: $\delta^* = (A^\top - I)^{-1} \pi(u^*)$ must be *integral* to host a state at all (the fence removes 20%, 55%, 85%, 97% of candidate data at $p = 1, 2, 3, 6$ in the censuses below), and nearest-vertex difference vectors have bounded Perron coordinate and bounded conjugates, hence form a finite set per specimen.

Lemma 4.3 (Norm floor). *An integral ℓ -fixed state of any power σ^p with $\delta^* \neq 0$ satisfies $|\langle \delta^*, L^{(m)} \rangle| \leq R'_m := B_m / (1 - |\beta_m|)^2$ uniformly in p (Dumont–Thomas prefix bound [5] over*

$|\beta_m^p - 1| \geq 1 - |\beta_m|$), and $|t^*| \geq \kappa(\sigma) := 1/(D^k \prod_{m \geq 2} R'_m) > 0$, where D clears the denominators of L , since the field norm of t^* is a nonzero rational of bounded denominator. In particular no genuine integral state has offset in $(0, \kappa)$: the small-offset regime is empty, and the integrality fence observed in the censuses is this floor in action.

Theorem 4.4 (Finite equivalence). *Let $\Delta(\sigma) = \{\delta \in \mathbb{Z}^k \setminus \{0\} : |\langle \delta, L \rangle| < L_{\max}, |\langle \delta, L^{(m)} \rangle| \leq R'_m\}$, a finite explicit set containing every ℓ - or r -fixed offset of every power. If $Q_\infty(\sigma)$ holds — every genuine state (i, δ, j) with $\delta \in \Delta(\sigma)$ contains an anchor or a coincidence in its (finite) descendant cone — then Conjecture 3.7 holds for σ ; conversely, failure of Q_∞ exhibits an anchor-free trapped component, refuting Conjecture 3.7. Thus $Q_\infty(\sigma)$ is equivalent to trapped anchoring, and the quantifier over powers in Theorem 4.2 is eliminated: Conjecture 3.7 is a single finite, effective check per substitution.*

Proof. ($\Rightarrow$) A trapped component contains an ℓ -cycle, ℓ -fixed for σ^p ; its offset is integral, conjugate-bounded and norm-floored by Lemma 4.3, hence in $\Delta(\sigma)$. Q_∞ places an anchor or coincidence in its cone, which lies in $C \cup \{\text{coincidences}\}$ by closedness; a coincidence contradicts $\text{leak} \equiv 0$. ($\Leftarrow$) If some Δ -state's cone avoids anchors and coincidences, the cone is finite (conjugates contract under inflation with bounded corrections), every state has a child, and a bottom SCC of its condensation has all children inside itself: closed, recurrent, with $\text{leak} \equiv 0$ and no anchor — an anchor-free trapped component. Note also that such a component is *totally cut-disjoint unconditionally*: by Lemma 3.5 with empty coincidence runs, any common interior cut would place a noncoincident aligned cell in it. $\square$

Per-specimen certificates (tail-corrected radius; an earlier under-converged radius computation was caught in audit and its outputs superseded, and a subsequent exact tail-corrected suffix radius—the superseding instrument—shrinks Δ a further $\sim 2.5\times$ with identical verdicts, e.g. flipped Tribonacci $|\Delta| = 402 \rightarrow 144$, plastic $2,504 \rightarrow 1,134$, re-certified states $2,868 \rightarrow 1,050$ and $17,460 \rightarrow 7,920$ respectively): Q_∞ is verified for the *complete exhaustive* $k = 3$, $|\sigma(a)| \leq 3$ corpus — all 4,554 primitive irreducible Pisot specimens of the 59,319-index enumeration, coverage confirmed by set equality against the directly enumerated PIP index family (a 47-index shard hole inside previously claimed coverage was exposed by exactly this reconciliation and gap-filled) — with 0 failures, 0 capped, 0 timeouts. The deepest specimens were closed by the optimal-slab and core certificates of §5 (largest retained closure 193,563 states; the former multi-million-state band collapses under the core slab). A further 24 random $k = 4$ specimens ($|\sigma(a)| \leq 3$, $|\beta_2|$ up to 0.976) certify with 0 failures, the five deepest via the core certificate of Theorem 5.4 (closures to 11.4 million states). Conjecture 3.7 therefore holds unconditionally for each certified specimen. Per-specimen decidability of pure discreteness is of course already given by [3]; the content here is the uniform target: the conjecture now reads “ $Q_\infty(\sigma)$ for all primitive irreducible Pisot σ ”, a statement about a norm-floor-bounded finite offset family, with no powers and no dynamics over p .

5 The saturated core

For $m \geq 2$ let B_m denote the one-step prefix-conjugate bound of Lemma 4.3,

$$B_m = \max_{a \in \mathcal{A}} \max_{p \text{ prefix of } \sigma(a)} |\langle \pi(p), L^{(m)} \rangle|,$$

and for $\varepsilon \geq 0$ define the *saturated core*

$$\mathcal{C}_\varepsilon(\sigma) = \left\{ \text{genuine } (i, \delta, j) : |\langle \delta, L^{(m)} \rangle| \leq \frac{2B_m}{1 - |\beta_m|} + \varepsilon \text{ for all } m \geq 2 \right\}.$$

Lemma 5.1 (Core invariance). $\mathcal{C}_\varepsilon$ is forward-invariant under overlap inflation for every $\varepsilon \geq 0$.

Proof. A child offset is $\delta' = A^\top \delta \pm (\text{pp difference})$, so $\langle \delta', L^{(m)} \rangle = \beta_m \langle \delta, L^{(m)} \rangle + \rho$ with $|\rho| \leq 2B_m$. Writing $F_m = 2B_m/(1 - |\beta_m|)$: if $|\langle \delta, L^{(m)} \rangle| \leq F_m + \varepsilon$ then $|\langle \delta', L^{(m)} \rangle| \leq |\beta_m|(F_m + \varepsilon) + 2B_m = F_m + |\beta_m|\varepsilon \leq F_m + \varepsilon$. $\square$

Lemma 5.2 (Entry). Let $\varepsilon > 0$ and let S be any family of genuine states with $|\langle \delta, L^{(m)} \rangle| \leq R_m$ for all m . Every descendant chain of every $s \in S$ enters $\mathcal{C}_\varepsilon$ within $h(\varepsilon) = \max_m \lceil \log((R_m - F_m)/\varepsilon) / \log(1/|\beta_m|) \rceil$ inflation levels.

Proof. Every genuine state has at least one child by Theorem 2.1 ($w(s) > 0$). Along any chain the recursion of Lemma 5.1 gives $|\langle \delta_n, L^{(m)} \rangle| - F_m \leq |\beta_m|^n (R_m - F_m)$. $\square$

Lemma 5.3 (Recurrent states are core). Every state lying on a directed cycle of the genuine overlap graph belongs to $\mathcal{C}_0$ (the closed ball; no ε).

Proof. If s lies on a cycle of length c , it is its own nc -step descendant for every n ; iterating the recursion around the cycle, $|\langle \delta_s, L^{(m)} \rangle| (1 - |\beta_m|^{nc}) \leq 2B_m (1 - |\beta_m|^{nc}) / (1 - |\beta_m|)$; divide. $\square$

Theorem 5.4 (Core reduction). For every $\varepsilon > 0$ the core certificate $Q_\infty^{\text{core}}(\sigma)$ — every state of $\mathcal{C}_\varepsilon(\sigma)$ contains an anchor or a coincidence in its descendant cone — is equivalent to trapped anchoring (Conjecture 3.7 for σ), and the cone may be searched inside $\mathcal{C}_\varepsilon$ alone: anchors and coincidences lie in $\mathcal{C}_0$ (for $\delta = 0$ the conjugates vanish; for $\delta = e_i - e_j$ they are bounded by $2B_m$), and $\mathcal{C}_\varepsilon$ is invariant, so the certificate is self-contained.

Proof. If a trapped component exists, its states are recurrent, hence in $\mathcal{C}_0 \subseteq \mathcal{C}_\varepsilon$ by Lemma 5.3; the core certificate places an anchor or coincidence in such a state's cone, which lies in $C \cup \{\text{coincidences}\}$ by closedness — either contradicts $\text{leak} \equiv 0$ exactly as in Theorem 4.4. Conversely, if some $\mathcal{C}_\varepsilon$ -state's cone avoids anchors and coincidences, the cone is finite (its conjugates are bounded by invariance and its Perron coordinates by the window), every state has a child, and a bottom SCC of its condensation is an anchor-free trapped component, as in Theorem 4.4. $\square$

Remark 5.5 (Optimal slab; norm-floor upgrade). Lemma 5.3 pins every cycle state — in particular the ℓ -cycle witness of any trapped component — to conjugate radius $2B_m/(1 - |\beta_m|)$, improving the bound of Lemma 4.3 from quadratic to linear in $(1 - |\beta_m|)^{-1}$ whenever $|\beta_m| > 1/2$, with the offset floor $\kappa(\sigma)$ rising correspondingly. Neither bound dominates globally ($B_m/(1 - |\beta_m|)^2$ is smaller for $|\beta_m| < 1/2$); the optimal certificate slab takes $\min(B_m/(1 - |\beta_m|)^2, 2B_m/(1 - |\beta_m|))$ per embedding. Per cycle, the pinned orbit of Lemma 3.4 satisfies the exact refinement $|\langle \delta_s, L^{(m)} \rangle| \leq 2B_m \sum_{i < c} |\beta_m|^i / |1 - \beta_m^c|$; for complex β_m away from resonance ($\arg \beta_m^c$ away from 0) this is $O(B_m c)$, and the Pisot property excludes exact resonance ($\prod_m (1 - \beta_m^n) \neq 0$, Lemma 3.4) but not near-resonance, so no uniform $O(B_m)$ bound follows; the near-resonant regime is the identified hard case. Conjecture 3.7 in its sharpest form now reads: every recurrent noncoincident state with conjugates at the one-step-saturated scale $2B_m/(1 - |\beta_m|)$ lies in a producing component.

Remark 5.6 (Computational verification; the core as enabling step). *Exact verification (no tolerance beyond 10^{-9} slab fattening): core invariance holds with zero escapees on all 29 computed core closures (5 at $k = 3$, 24 at $k = 4$); on the largest $k = 3$ specimen the closed-ball bound is realized to four decimals (maximal conjugate 39.1870 against radius 39.1880). Lemma 5.3 verified on all 24 evaluable closures ($k = 3$ and $k = 4$): all recurrent states inside the core in every embedding. The recurrent skeleton is consistently minuscule (0.01–2.9% of the closure; 557 of 600,357 states on a representative deep specimen). The reduction factor of the core certificate scales like $(1 - |\beta_2|)^{-2}$ and reaches $87\times$ at $k = 3$, $|\beta_2| = 0.94$; at $k = 4$ it is an enabling step rather than an optimization — the R' -slab certificate exhausts a 4 GB working set at $|\beta_2| \approx 0.946$ while the core certificate completes at $|\beta_2| = 0.976$ with an 11,434,672-state closure. As throughout, these verifications are elimination, not validation: no census bears on the uniform statement.*

6 Powers, period, and the trapped shape

Lemma 6.1 (Pisot field identity). *For every Pisot number β and every $p \geq 1$, $\mathbb{Q}(\beta^p) = \mathbb{Q}(\beta)$. Consequently the k values β_m^p are pairwise distinct, and the characteristic polynomial of A^p is the (irreducible) minimal polynomial of β^p .*

Proof. If an embedding σ fixes β^p but moves β , then $\sigma(\beta) = \beta\zeta$ with $\zeta^p = 1$, $\zeta \neq 1$, so $|\sigma(\beta)| = \beta > 1$, contradicting the Pisot property. Hence $\deg \beta^p = k$, so the Galois orbit of β^p has k distinct elements; these lie among the k values β_m^p , which are therefore pairwise distinct, and $\chi_{A^p} = \prod_m (x - \beta_m^p)$ is the minimal polynomial of β^p . $\square$

Corollary 6.2 (Power closure of the PIP class). *If σ is primitive irreducible Pisot, so is σ^p for every $p \geq 1$. Moreover, when β has a complex conjugate pair, $\arg \beta_2 \notin \pi\mathbb{Q}$: a rational angle would give $\beta_2^{2b} = \beta_3^{2b}$ for some b , collapsing the Galois orbit of β^{2b} below degree k .*

Theorem 6.3 (Period-scaled size floor). *Let C be a closed recurrent noncoincident component with leak $\equiv 0$ and period p , with the integer intertwining $N_C V = V A$, V injective, of Theorem 2.5. Write $V_t = P_t V$ for the coordinate projection P_t onto the cyclic class C_t . Then $N_C V_t = V_{t+1} A$ (indices mod p), every V_t is injective, $|C_t| \geq k$ for every t , and $|C| \geq pk$. The mass-balance eigen-relation refines classwise: $N_C w_t = \beta w_{t+1}$ with $w_t = V_t L > 0$.*

Proof. N_C maps class t into class $t + 1$, so projecting $N_C V = V A$ gives $N_C V_t = V_{t+1} A$. If some $V_t = 0$ then $V_{t+1} A = 0$ and A is invertible, so $V_{t+1} = 0$; around the cycle $V = 0$, a contradiction. Composing p steps, $N_C^p V_t = V_t A^p$, so $\ker V_t$ is A^p -invariant. By Lemma 6.1, $\mathbb{Q}[A^p]$ acts on $\mathbb{Q}^k \cong \mathbb{Q}(\beta)$ as multiplication by β^p with $\mathbb{Q}(\beta^p) = \mathbb{Q}(\beta)$, so $\mathbb{Q}^k$ is a simple $\mathbb{Q}[A^p]$ -module; $V_t \neq 0$ forces V_t injective. $\square$

Theorem 6.4 (Power reduction; aperiodicity without loss of generality). *Let C be a trapped component of period p for σ . Then each cyclic class C_t is a trapped component for σ^p — closed (all p -step descendants of C_t remain in C_t), recurrent with primitive internal matrix $B_t = N_C^p|_{C_t}$, noncoincident, leak $\equiv 0$, and anchor-free if C is. Since σ^p is again primitive irreducible Pisot (Corollary 6.2), Conjecture 3.7 for all primitive irreducible Pisot substitutions is equivalent to its restriction to trapped components with primitive (aperiodic) internal matrix. Every attack on, or hunt for a counterexample to, uniform trapped anchoring may therefore assume N_C primitive.*

Proof. Closedness with leak $\equiv 0$ means every one-step descendant of C lies in C ; p -step descendants of C_t lie in $C_{t+p} = C_t$. B_t is irreducible and aperiodic by the Perron–Frobenius

cyclic-class theory; the remaining properties are inherited state-by-state, and the anchor set $\{\delta = 0\} \cup \{\delta = e_i - e_j\}$ does not depend on the substitution. Conversely, an aperiodic trapped component is a trapped component, so the restricted conjecture implies nothing weaker. $\square$

Theorem 6.5 (Frozen rays). *Up to replacing σ by a power (preserving the primitive irreducible Pisot class by Corollary 6.2 and trappedness with primitive N_C by Theorem 6.4), every trapped component contains an ℓ -fixed state s_ℓ^* and an r -fixed state s_r^* , both with $t^* \neq 0$. For each, the associated pair of σ -fixed half-tilings at offset t^* (the setting of Lemma 4.1) is cut-disjoint along the entire ray: no level ever produces a common cut.*

Proof. The leftmost-cell map ℓ is a self-map of C (closedness with leak $\equiv 0$), hence has a cycle of some length q_ℓ ; a cycle state is ℓ -fixed for σ^{q_ℓ} , and likewise for r . Pass to σ^q , $q = \text{lcm}(q_\ell, q_r)$: this is again primitive irreducible Pisot, C is again trapped with N_C^q primitive, and the chosen cycle states are ℓ - and r -fixed. Anchor-freeness excludes $\delta = 0$, and $\langle \delta, L \rangle = 0$ iff $\delta = 0$ by linear independence of the L_a , so $t^* \neq 0$. A common cut at any refinement level places, by Lemma 3.5, an aligned (anchor) state in C , contradicting anchor-freeness. $\square$

Remark 6.6 (Honest placement). *Theorem 6.5 localizes the open core to two concrete infinite objects per hypothetical counterexample: two pairs of σ -fixed rays, offset by core-bounded t_ℓ^*, t_r^* , whose cut sets are disjoint forever — equivalently, whose prefix-Parikh difference walk, confined to the finite saturated core of §5, avoids the fixed lattice point δ^* at every step. This is Lee’s algebraic coincidence localized as in Lemma 4.1, now with the avoidance walk’s range pinned to the core and the substitution normalized to be aperiodic with ℓ - and r -fixed witnesses. It is a sharpening of the counterexample shape, not progress on the avoidance question itself: forever avoiding one point of a finite visited set is a positional constraint that no count-level or marginal argument can refute.*

Theorem 6.7 (Transversal substitution). *Let C be a trapped component, normalized as in Theorem 6.5 (N_C primitive, s_ℓ^* ℓ -fixed). The map $\Sigma_C: s \mapsto (\text{children of } s \text{ in left-to-right order})$ is a well-defined substitution on the finite alphabet C (trappedness excludes coincidence letters), with abelianization N_C , primitive, admitting the one-sided fixed point $\tau = \lim \Sigma_C^n(s_\ell^*)$; and τ is exactly the transversal sequence of the frozen ray pair of Theorem 6.5 — the left-to-right sequence of overlap cells of the two rays’ common refinement, every cell of which lies in C . The natural tile lengths $w(s)$ realize τ as a self-similar tiling of the ray with inflation β . Consequently a trapped component is equivalent to the following package: a primitive substitution Σ on $|C| \geq k$ letters with reducible characteristic polynomial $\chi_A \cdot q(x)$ and Pisot–Perron expansion β of algebraic degree $k < |C|$, equipped with the integer valuation $s \mapsto v_s$ intertwining N_C with A (Theorem 2.5), whose fixed point never meets the coincidence diagonal.*

Proof. Children of a state, being the cells of an interval partition of $\beta \cdot o(s)$, carry a canonical left-to-right order, and by closedness with leak $\equiv 0$ all lie in C , so Σ_C is a substitution over C with abelianization N_C , primitive by normalization. ℓ -fixedness says $\Sigma_C(s_\ell^*)$ begins with s_ℓ^* , so the one-sided fixed point τ exists. Each cell of the rays’ common refinement occurs, suitably translated, in every sufficiently deep refinement of s_ℓ^* along the tracked anchoring of Lemma 4.1, hence lies in C , and the level- n cell sequences are by construction $\Sigma_C^n(s_\ell^*)$; pass to the limit. The width vector w is the Perron eigenvector of N_C (Theorem 2.5), giving the self-similar realization with inflation β . The degree statement is Lemma 6.1; injectivity of V gives the intertwining. $\square$

Remark 6.8 (Placement and contact surface). *Theorem 6.7 moves the counterexample shape into a classical-looking category: primitive substitutions with reducible Pisot–Perron expansion, fibered over an irreducible Pisot substitution, with a diagonal-avoidance property along the fixed point. The avoidance itself remains positional — it is the cut-disjointness of Theorem 6.5 re-expressed — but the package now sits where the reducible-Pisot and Property (F) / bounded-remainder literatures have tools, and where any rigidity statement for reducible expansions with irreducible Pisot degree would bear directly on Conjecture 3.7.*

Lemma 6.9 (Unit step). *Let s_1, s_2 be consecutive cells (left to right) of the common refinement of two tile grids, with $s_1 = (i_1, \delta_1, j_1)$, $s_2 = (i_2, \delta_2, j_2)$, and suppose their common boundary is not a shared cut of the two grids. Then exactly one letter changes, and*

$$\delta_2 - \delta_1 = \begin{cases} -e_{i_1}, & j_2 = j_1 \text{ (an } I\text{-grid cut: the } i_1\text{-tile ends),} \\ +e_{j_1}, & i_2 = i_1 \text{ (a } J\text{-grid cut: the } j_1\text{-tile ends).} \end{cases}$$

At a shared cut both letters change and $\delta_2 - \delta_1 = e_{j_1} - e_{i_1}$ (the double step), and the configuration produces aligned cells as in Lemma 3.5.

Proof. Crossing an I -cut increments the prefix-Parikh vector of the I -grid by e_{i_1} , which enters the offset with a minus sign; a J -cut enters with a plus sign; both formulas are immediate from $\delta = A^\top \delta_{\text{parent}} + \pi(\text{pp}_J) - \pi(\text{pp}_I)$ read across one cut. A simultaneous cut composes the two. (Machine validation: 1,148 consecutive-cell boundaries across three certified specimens; the rule holds at every boundary that is not a shared cut, and all six exceptions decode exactly as shared-cut configurations adjacent to coincidence cells.) $\square$

Corollary 6.10 (The avoidance walk is a substitutive unit-step walk). *For a trapped component, normalized as in Theorem 6.5, the offset sequence (δ_n) along the transversal fixed point τ of Theorem 6.7 is a walk with steps in $\{-e_a\} \cup \{+e_b\}$ — the sign recording which ray’s tile ends at the n -th cut, the letter recording which tile — with no double steps (cut-disjointness), confined to the closed saturated core C_0 , generated by the primitive substitution Σ_C , and avoiding $\{0\} \cup \{e_i - e_j\}$ at every step, forever. The open core of the conjecture, in its most concrete form to date: such a substitutive unit-step walk in a finite region with these forbidden points cannot exist. Its impossibility remains positional — the walk simply never lists the forbidden points among its finitely many visited values — and is untouched here.*

Lemma 6.11 (Type preservation; fibering over the Barge–Diamond complex). *In a trapped transversal, call a cell boundary type I or type J according to which ray’s grid has a vertex there (exactly one does, by cut-disjointness). The substitution dynamics on transitions — sending the boundary between τ_n, τ_{n+1} to the boundary between the last child of τ_n and the first child of τ_{n+1} — preserves type. On type- I transitions it acts on the underlying I -letter pair by $(i, i') \mapsto (\text{last } \sigma(i), \text{first } \sigma(i'))$, i.e. it fibers over the classical Barge–Diamond two-letter dynamics of σ ; symmetrically for type J .*

Proof. The boundary at x maps to the boundary at βx under the self-similarity of the frozen pair. If x is an I -vertex, βx is a vertex of the refined I -grid; it cannot also be a vertex of the refined J -grid, for that would be a shared cut of the pair — which is self-similar, hence cut-disjoint at every level (Theorem 6.5). The I -letters at the image boundary are the last letter of $\sigma(i)$ (the refined I -tile ending at βx) and the first letter of $\sigma(i')$, while the ambient J -letter is unchanged across βx . $\square$

Remark 6.12 (Localization of the homological-Pisot question for Σ_C). By Lemma 6.11, the transition subcomplex of Σ_C decomposes edge-wise into an I -part and a J -part, each carrying dynamics fibered over the Barge–Diamond complex of σ itself (whose eventual-range structure, for irreducible σ without asymptotic cycles, is the classical loop-free object). Whether Σ_C is homological Pisot — which by the structural fact of Barge–Bruin–Jones–Sadun would force every eigenvalue of N_C outside χ_A to be zero or a root of unity, a spectral shape realized by no recurrent component in the certified corpus — therefore reduces to: (1) the spectral condition on N_C itself, (2) the component count of the eventual range against the multiplicity of the eigenvalue 1, and (3) loop-freeness of the eventual range, now a question about cycles in a complex fibered over a known loop-free base, settled by the orientation/reducedness argument of Lemma 7.6. Section 7 resolves (3) affirmatively over loop-free bases (Theorem 7.7) and converts (2) into an unconditional spectral constraint (Corollary 7.5); the residue of the rigidity question is the spectral condition (1) in the sharpened form of Corollary 7.8.

Remark 6.13. Computational sanity (elimination-grade): $\deg \beta^p = 3$ verified for sampled specimens across $p \in \{2, 3, 6, 12, 24\}$, and the full PIP closure of σ^p (primitivity, irreducibility, distinctness of β_m^p) for six specimens across $p \in \{2, 3, 4, 6, 8, 12\}$. Corollary 6.2 explains a feature of the recurrent-cycle census: the resonance quantity $|1 - \beta_2^c|$ over realized cycle lengths is bounded below by 0.347 across 3,145 certified specimens and never approaches 0 through angular near-resonance — as it cannot exactly, $\arg \beta_2$ being an irrational angle. The hypothetical counterexample to Conjecture 3.7 is now simultaneously constrained in size ($|C| \geq pk$, and WLOG $p = 1$ with N_C primitive), location (the saturated core of §5), spectrum ($\rho(N_C) = \beta$ exactly, $\chi_A \mid \chi_{N_C}$), and geometry (totally cut-disjoint at all scales).

7 The transition complex of Σ_C : decomposition, inherited loop-freeness, and the coincidence rank of the pair system

Throughout this section C is a trapped component, normalized as in Theorem 6.7, and — except in Lemma 7.10 and its corollaries, which need no such hypothesis — C is *anchor-free* (the H3'-counterexample shape), so that by Theorem 6.5 no state has offset 0 or $L_i - L_j$ and the transversal is totally cut-disjoint.

We use the exact transition-complex conventions of Barge–Diamond [11] as stated in Barge–Bruin–Jones–Sadun [12]: for a primitive aperiodic substitution φ on an alphabet $\mathcal{B}$, the complex G_0 has one edge v_{ab} per legal two-letter word ab , running from the vertex out_a (the end of the letter edge e_a) to the vertex in_b (the beginning of e_b); since the letter edges are not part of G_0 , the vertices out_a and in_a are *distinct* and G_0 is bipartite. Substitution acts by $v_{ab} \mapsto v_{\text{last } \varphi(a), \text{first } \varphi(b)}$ and permutes the edges of the eventual range G_0^{ER} . With $b_1 = E - V + C$ the first Betti number of G_0^{ER} and C its component count, the exact sequence of [11, 12] (for a power fixing the ER edges, whose image of $\tilde{H}^0(G_0^{ER})$ lies in the +1 eigenspace of the transposed abelianization) gives $\dim \tilde{H}^1(\Omega_\varphi, \mathbb{Q}) = \dim \varinjlim M_\varphi^\top - (C - 1) + b_1$.

Lemma 7.1 (Connectivity for irreducible Pisot). *If φ is irreducible Pisot of degree k then $G_0^{ER}(\varphi)$ is connected and $\dim \tilde{H}^1(\Omega_\varphi) = k + b_1(G_0^{ER})$; φ is homological Pisot iff $b_1 = 0$ iff φ has no asymptotic cycles.*

Proof. 1 is not an eigenvalue of any power of the abelianization: the eigenvalues of the q -th power are β^q and β_m^q with $\beta^q > 1$ and $|\beta_m| < 1$. So $\tilde{H}^0(G_0^{ER})$ injects into the trivial +1 eigenspace, forcing $C = 1$, and $\dim \varinjlim A^\top = k$ since $\det A \neq 0$. $\square$

(Machine validation: $\mathcal{C} = 1$ on all 4,554 exhaustive $k = 3$ PIP specimens and 1,190 random $k = 4$ specimens, exact-convention instrument gated against the calibration examples of [12], including the asymptotic-cycle specimen $1 \mapsto 21112$, $2 \mapsto 121$ with $b_1 = 1$ and the coincidence-rank-3 triple cover with $\mathcal{C} = 3$, $b_1 = 0$.)

Remark 7.2. *The integers $\mathcal{C}$ and b_1 are invariants of the pair $(\Omega, \text{chosen Barge–Diamond complex})$, not of Ω alone: a collared presentation can change both while preserving $\check{H}^1(\Omega)$ (e.g. the uncollared Fibonacci S_0^{ER} is contractible with $\mathcal{C} = 1$, a collared one has $\mathcal{C} = 2$; both give $\check{H}^1 = \mathbb{Z}^2$). We fix the uncollared one-edge-per-legal- two-word convention of [11, 12] throughout, so all reported b_1 are mutually comparable; every statement of this section uses $\mathcal{C}$ and b_1 of that fixed complex and is set against the homeomorphism invariant $\dim \check{H}^1$ (cf. the general 1-dimensional sequence as recorded in [14]).*

Lemma 7.3 (Boundary-type determinacy). *Let $c = (i, \delta, j)$ be any genuine overlap with $t = \langle \delta, L \rangle$, so $o(c) = [\max(0, t), \min(L_i, t + L_j))$. The left endpoint of $o(c)$ is a vertex of the I -grid iff $t < 0$, of the J -grid iff $t > 0$, and of both iff $t = 0$ (the prefix anchor); the right endpoint is a vertex of the I -grid iff $t > L_i - L_j$, of the J -grid iff $t < L_i - L_j$, and of both iff $t = L_i - L_j$ (the suffix anchor). In particular, for an anchor-free trapped component every cell has a well-defined left type and right type, intrinsic to its state, and the type of the boundary between consecutive transversal cells equals the right type of the left cell and the left type of the right cell.*

Proof. Immediate from $o(c) = [0, L_i) \cap [t, t + L_j)$: the left endpoint is 0 (an I -tile vertex, interior to the J -tile) when $t < 0$ and t (a J -tile vertex, interior to the I -tile) when $t > 0$; symmetrically on the right. By irreducibility, $t = 0$ iff $\delta = 0$ and $t = L_i - L_j$ iff $\delta = e_i - e_j$, the two anchors. Consistency across a boundary is Lemma 6.9 read geometrically. (Machine validation: 9,945 consecutive-cell boundaries across 24 PIP specimens, exact integer-vector cut positions; the sign predicates match the geometric type at all 9,660 non-shared boundaries, and all 285 shared cuts are anchor-adjacent.) $\square$

Theorem 7.4 (Type decomposition). *For an anchor-free trapped C , every edge of $G_0(\Sigma_C)$ into in_t has type equal to the left type of t , and every edge out of out_s has type equal to the right type of s . Consequently $G_0(\Sigma_C)$, and hence $G_0^{ER}(\Sigma_C)$, is the vertex-disjoint union of its type- I part G_I and type- J part G_J ; in particular $b_1(G_0^{ER}(\Sigma_C)) = b_1(G_I) + b_1(G_J)$. Both parts meet the eventual range, so*

$$\mathcal{C}(G_0^{ER}(\Sigma_C)) \geq 2.$$

Proof. The first sentence is Lemma 7.3: the type of an edge (s, t) is the type of the shared boundary, which is determined by either incident state. A vertex is therefore incident to edges of one type only, and an edge joins an out-vertex and an in-vertex of its own type: the two parts share no vertices. Both parts are nonempty in the eventual range: type- I and type- J legal two-words exist at every level (each ray's tiles are finite, so each grid cuts the transversal syndetically), the substitution map preserves type (Lemma 6.11), and the eventual range is the image of the full legal set under a high power. $\square$

Corollary 7.5 (Root-of-unity spectrum; size-floor upgrade). *Write $\chi_{N_C} = \chi_A \cdot q(x)$ (Theorem 2.5). For an anchor-free trapped C , at least $\mathcal{C}(G_0^{ER}(\Sigma_C)) - 1 \geq 1$ roots of q , counted with multiplicity, are roots of unity. In particular $\deg q \geq 1$, so $|C| \geq k + 1$, strictly improving Theorem 2.5; and $\text{spec}(N_C)$ must contain a root of unity. Containment of a root of unity is not by itself rare — it occurs in 225 of 358 productive recurrent components of the certified corpus.*

The trapped fingerprint is the exhaustive clause of Corollary 7.8, that every nonzero root of q be a root of unity (equivalently $\dim \check{H}^1(\Omega_{\Sigma_C}) = k$); this is realized by no productive component, 226 of 245 sampled carrying an off-unit-circle Perron root $\rho(N_C) > 1$.

Proof. In the exact sequence for the edge-fixing power Σ_C^q , the group $\tilde{H}^0(G_0^{ER}) \cong \mathbb{Q}^{C-1}$ injects into the fixed space of the limit automorphism, so the eigenvalue 1 of N_C^q has geometric multiplicity at least $C - 1$; each such eigenvalue comes from a root ζ of χ_{N_C} with $\zeta^q = 1$, and no root of χ_A is a root of unity (Pisot). Theorem 7.4 gives $C \geq 2$. $\square$

Lemma 7.6 (Reduced projection). *Every simple cycle of $G_0^{ER}(\Sigma_C)$ is type-pure (Theorem 7.4) and projects, via the letter coordinate of its type ($s \mapsto i(s)$ for type I , $s \mapsto j(s)$ for type J), to a reduced closed edge-path in $G_0^{ER}(\sigma)$; hence its existence forces $b_1(G_0^{ER}(\sigma)) \geq 1$.*

Proof. Take the cycle type- I (type J is symmetric in the other coordinate). A type- I edge (s, t) projects to the legal σ -two-word $(i(s), i(t))$ of consecutive I -tiles; by Lemma 6.11 the projection intertwines the two-word dynamics, and since ER edges lie on cycles of the edge permutation, the projected edge lies on a cycle of σ 's two-word map, hence in $G_0^{ER}(\sigma)$. By Lemma 6.9, a type- I predecessor of t is determined by its i -letter ($s = (i(s), \delta_t + e_{i(s)}, j(t))$) and a type- I successor of s by its i -letter; so two distinct cycle edges sharing in_t (resp. out_s) project to *distinct* base edges sharing $\text{in}_{i(t)}$ (resp. $\text{out}_{i(s)}$). Consecutive edges of the alternating chain always share an in- or out-vertex, so the projected closed walk never traverses a base edge and immediately retraverses it: it is reduced in the free-group sense (the only relevant one, since edge reduction is a local, consecutive-pair operation; non-consecutive coincidences of projected vertices are irrelevant). Each type- I edge advances the i -coordinate by one I -tile (Lemma 6.9), so the walk has positive length. A reduced nontrivial closed edge-walk represents a nontrivial element of $\pi_1(G_0^{ER}(\sigma))$; a graph with nontrivial fundamental group is not a tree, so $b_1(G_0^{ER}(\sigma)) \geq 1$. (The argument is through π_1 , not the homology class of the projected loop itself, which may vanish in H_1 ; existence of *any* reduced nontrivial closed walk already forces a cycle in the graph.) $\square$

Theorem 7.7 (Inherited loop-freeness). *If σ has no asymptotic cycles ($b_1(G_0^{ER}(\sigma)) = 0$); equivalently, by Lemma 7.1, $\dim \check{H}^1(\Omega_\sigma) = k$), then for every anchor-free trapped component C of σ ,*

$$b_1(G_0^{ER}(\Sigma_C)) = 0.$$

Condition (3) of Remark 6.12 is therefore automatic over loop-free bases.

Proof. $b_1 > 0$ produces a simple cycle; Lemma 7.6 contradicts $b_1(G_0^{ER}(\sigma)) = 0$. $\square$

Corollary 7.8 (The Q-A residue). *Over a loop-free base, Σ_C is homological Pisot if and only if the number of nonzero roots of q , with multiplicity, equals $C(G_0^{ER}(\Sigma_C)) - 1$; in that case every nonzero root of q is a root of unity. The rigidity question for the trapped shape is thus exactly a counting identity between the extra spectrum of N_C and the components of its transition complex.*

Remark 7.9 (The residue is irreducible to the base loop-freeness). *On the loop-free base stratum Theorem 7.7 supplies $b_1(G_0^{ER}(\Sigma_C)) = 0$, and Corollary 7.5 supplies the lower half of Corollary 7.8: at least $C - 1$ of the nonzero roots of q are roots of unity. The entire residue is then the matching upper bound, that the number r of nonzero roots of q does not exceed $C - 1$. This upper bound is not a consequence of loop-freeness: for a producing within-SCC matrix the nonzero spectrum is essentially full (across 152 recurrent components the ratio of nonzero spectrum to size has median 1.000, and only 7.9% have at most $k - 1$ nonzero eigenvalues), so $r \leq C - 1$ is*

a strongly non-generic spectral collapse — precisely the homological-Pisot hypothesis itself, not a corollary of $b_1 = 0$. Equivalently, for connected $G_0^{ER}(\Sigma_C)$ it demands $q(x) = x^{|C|-k}$ (all extra spectrum nilpotent), the spectrally minimal trapped shape. The identity $r = (C - 1) - b_1$ and the root-of-unity clause are confirmed on the two homological-Pisot triple covers of [12]: there $q = x^a(x - 1)^2$ with $C = 3$, $b_1 = 0$, $r = 2$.

Proof. $\dim \check{H}^1(\Omega_{\Sigma_C}) = \dim \varinjlim N_C^\top - (C - 1) + b_1 = (k + \#\{\text{nonzero roots of } q\}) - (C - 1)$ by Theorem 7.7; set this equal to $k = \deg \beta$. Equality forces all nonzero roots of q to be roots of unity by Corollary 7.5. $\square$

Lemma 7.10 (Leg factor). *For any trapped component C (no anchor-freeness needed), $(\Omega_{\Sigma_C}, \mathbb{R})$ factors onto $(\Omega_\sigma, \mathbb{R})$.*

Proof. A tiling $S \in \Omega_{\Sigma_C}$ assigns to $\mathbb{R}$ a bi-infinite cell sequence with widths $w(s)$; merge maximal runs of cells between consecutive I -grid vertices (well defined by Lemma 7.3, shared or not) and label each merged tile by the common i -letter. I -vertices are syndetic (each I -tile has length at most $L_{\max}$ and cells have width at least $\min_s w(s) > 0$), every patch of S is a translate of a patch of τ , where the merged tiles are precisely the I -grid tiles of the frozen ray of length L_i , so the merged object is locally allowed for σ and $\pi_1(S) \in \Omega_\sigma$. The local rule π_1 is continuous and translation-equivariant; its image is a nonempty closed invariant subset of the minimal Ω_σ , hence everything. $\square$

Corollary 7.11 (Coincidence rank of the pair system). *$\text{cr}(\Sigma_C) \geq 2$ for every trapped component. If moreover σ is unimodular and Σ_C is homological Pisot, then $\text{cr}(\Sigma_C) \geq 3$; and if $\text{cr}(\Sigma_C) = 2$ (for unimodular σ) then $\dim \check{H}^1(\Omega_{\Sigma_C}) \geq 2k - 1$.*

Proof. Σ_C is a primitive, non-periodic, FLC substitution on letters with Pisot dilatation β (flow-periodicity of a transversal would push to flow-periodicity of $\pi_1(S)$, impossible), so $\text{cr}(\Sigma_C) = 1$ iff Ω_{Σ_C} has pure discrete spectrum [13, Thm. 4(5)]. Pure discreteness passes to factors, and Ω_σ is a factor (Lemma 7.10); but σ carries a trapped component, hence fails overlap coincidence and pure discreteness [10, 3]. So $\text{cr}(\Sigma_C) \geq 2$. (The trapped component here is the *overlap-graph* object, for which overlap coincidence $\Leftrightarrow$ pure discreteness is a genuine equivalence. Its identification with the balanced-pair closed nonproductive SCC of the main manuscript is the “bridge converse” made explicit there (PSC-PROOF v15, §6.5): balanced-pair non-termination implies overlap non-coincidence only through that identification, which the main manuscript therefore states as a hypothesis.) The norm of β is $\pm \det A = \pm 1$, odd; if $\text{cr}(\Sigma_C) = 2$ then $\dim \check{H}^1(\Omega_{\Sigma_C}) \geq 2k - 1$ by [13, Thm. 22] (whose hypotheses are exactly: Pisot substitution on letters, odd norm, coincidence rank two — reducibility of Σ_C is permitted), which exceeds k for $k \geq 2$, contradicting homological-Pisotness. $\square$

Theorem 7.12 (Conditional placement: the unimodular conjecture in the topological orbit). *Let Q -A denote the statement that Σ_C is homological Pisot for every trapped component of every unimodular irreducible Pisot substitution, and CRC the Coincidence Rank Conjecture of [12, 13]. Then*

$$Q\text{-}A + \text{CRC} \implies \text{unimodular PSC}.$$

Proof. Suppose σ is unimodular irreducible Pisot without pure discrete spectrum. Failure of overlap coincidence produces a closed recurrent noncoincident component with $\rho(N_C) = \beta$ and $\text{leak} \equiv 0$ (Theorem 2.2 with the Akiyama–Lee criterion [3]; cf. Remark 3.9), i.e. a trapped

component, which persists under the power normalization of Theorem 6.4 (pure discreteness is power-invariant). By Q-A, $\dim \check{H}^1(\Omega_{\Sigma_C}) = \deg \beta$, and the norm of β is ± 1 , so CRC gives $\text{cr}(\Sigma_C) \mid 1$, i.e. pure discreteness of Ω_{Σ_C} [13, Thm. 4(5)], hence of its factor Ω_σ (Lemma 7.10) — contradicting the hypothesis. $\square$

Remark 7.13 (Honest placement; adjudication of the branching question). (i) *CRC is open precisely in the regime $\text{cr} \geq 3$ that Corollary 7.11 lands in (it is proved for degree one [12] and for coincidence rank two [13]), and Q-A is open; Theorem 7.12 transfers the unimodular conjecture into the orbit of the homological-Pisot/CRC program rather than resolving anything.* (ii) *All results of this section are count- and topology-level constraints on the hypothetical trapped shape; the positional avoidance residue of Corollary 6.10 is untouched, as the no-coarse-quotient obstruction requires.* (iii) *A formulation of the loop question as “backward branching of the avoidance walk” (two trapped pasts of one tail) has no content: asymptotic pairs exist in every primitive aperiodic substitution subshift, every point of the hull of τ decodes to a trapped pair, and recognizability of Σ_C constrains desubstitution, not backward determinism of the shift; the refutable object is exactly $b_1(G_0^{ER}(\Sigma_C))$ — by Theorem 7.4 a type-pure alternating cycle of asymptotic branchings — and over loop-free bases it is now excluded.* (iv) *Barge’s dichotomy [13, Cor. 25] (a PSC counterexample has an asymptotic cycle or coincidence rank at least three) calibrates the base-level census: in the exhaustive $k = 3$, $|\sigma(a)| \leq 3$ corpus, 3,102 of 4,554 specimens (including 1,740 of the 2,628 unimodular ones) are loop-free, hence homological Pisot with the $\text{cr} \geq 3$ alternative; the $b_1 > 0$ stratum, exactly quantified ($b_1 \in \{1, 2, 4\}$ at $k = 3$; b_1 up to the complete bipartite ceiling at $k = 4$, where $b_1 = 3$ also occurs), is where the asymptotic-cycle alternative lives. As always, no census bears on the uniform statements.*

8 Computational evidence (elimination, not validation)

All computations use the validated instrument (`overlap_residual.py`; partition identity verified to 5.6×10^{-16} , per-SCC restriction to 4.8×10^{-14} , gap formula to 2.1×10^{-14} including an SCC of size 5068).

corpus	closed SCCs	nonproducers	H3 violators	
$k = 3$, $ \sigma(a) \leq 3$, exhaustive (4,554 PIP)	1,386	0	0	zero-free $\Rightarrow$ suffix-align
$k = 3$, $ \sigma(a) \leq 4$, random	105	0	0	
$k = 3$, $ \sigma(a) \leq 6$, random	35	0	0	12 cap
$k = 4$, $ \sigma(a) \leq 4$, random	~ 60	0	0	
$k = 5$, $ \sigma(a) \leq 3$, random	12	0	0	

$Q(p)$ census (abstract states; no reachability or pure-discreteness input): collisions in 3,216/3,216 cases over $p \in \{1, 2, 3, 6\}$, with maximal cone depth 6, 3, 3, 1 respectively — the small-offset regime $t^* = c/(\beta^p - 1) \rightarrow 0^+$ is the easy one, and the difficulty of $Q(p)$ concentrates at small p (sampled pattern, $k = 3$).

Both anchor-only classes occur (prefix-only and suffix-only closed SCCs exist), so the disjunction in Theorem 3.6 cannot be strengthened to a single anchor type. By Theorem 3.6 the anchoring counts in the table are implied by the production counts; they are therefore consistency checks, and *no census evidence bears directly on Conjecture 3.7* — necessarily, since a

trapped specimen would refute the Pisot conjecture. Sampled minima/maxima are not proved bounds; capped specimens are inconclusive and excluded; image length is a cap throughout.

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